

The extra $(x + 1)$ factor: a proof of Karttunen's second conjecture on the binary columns of the Fibonacci numbers

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Abstract

OEIS A036284 reads one period of the n -th binary column of the Fibonacci numbers as a polynomial over $\text{GF}(2)$. A. Karttunen conjectured in November 1998 that this polynomial is divisible by $(x^3 + 1)^{2^{n-1}-1}$ (Conjecture 1) and that “there is also one extra $(x^1 + 1)$ factor present” (Conjecture 2). We prove Conjecture 2, that is that $(x^3 + 1)^{2^{n-1}-1}(x + 1)$ divides the n -th term. Equivalently, the $(x + 1)$ -adic valuation of the column polynomial is at least 2^{n-1} , one more than the valuation of $(x^3 + 1)^{2^{n-1}-1}$. The proof is a carry-periodicity identity read on the factor $x + 1$ instead of on $x^2 + x + 1$; the point is simply that $x + 1$ does not divide $x^2 + x + 1$, so the cancellation that costs one power on the $x^2 + x + 1$ side costs nothing here. The note is self-contained: the identity is stated and proved in Section 3.

1 The conjecture

Let $F(0) = 0$, $F(1) = 1$, $F(k) = F(k-1) + F(k-2)$. The bits in the n -th column from the right of the binary expansions of $F(0), F(1), F(2), \dots$ repeat with period $3 \cdot 2^n$, and OEIS A036284 (A. Karttunen, 1 November 1998, *Periodic vertical binary vectors of Fibonacci numbers*) records one period read as a binary number:

$$a(n) = \sum_{k=0}^{3 \cdot 2^n - 1} \left(\lfloor F(k)/2^n \rfloor \bmod 2 \right) 2^k, \quad a(0), \dots, a(3) = 6, 24, 1440, 5728448.$$

Identifying an integer with the $\text{GF}(2)$ polynomial whose coefficient of x^k is bit k , the entry carries

Conjecture: For $n \geq 1$, each term $a(n)$, when considered as a $\text{GF}(2)[X]$ -polynomial, is divisible by the $\text{GF}(2)[X]$ -polynomial $(x^3 + 1)^{A000225(n-1)}$ **Conjecture 2:** there is also one extra $(x^1 + 1)$ factor present, see A136384.

with $A000225(n-1) = 2^{n-1} - 1$. Here we prove Conjecture 2, in the form

$$(x^3 + 1)^{2^{n-1}-1}(x + 1) \mid a(n) \quad (n \geq 1).$$

As of the “Last modified” line on the live entry (23 August 2026) both statements are still recorded as unproven conjectures and no proof appears in the entry’s links or programs.

2 Notation

Fix $n \geq 1$ and set

$$L = 3 \cdot 2^n, \quad q(x) = x^2 + x + 1, \quad x^3 + 1 = (x + 1)q(x).$$

Note that $q(1) = 1$, so $x + 1 \nmid q$; this small fact is the whole difference between the two conjectures. For nonzero $P \in \text{GF}(2)[x]$ and irreducible f , let $v_f(P)$ be the exponent of f in P . By the Frobenius endomorphism,

$$x^{3 \cdot 2^m} + 1 = (x^3 + 1)^{2^m} \quad (m \geq 0). \quad (1)$$

Let $r_k = F(k) \bmod 2^{n+1}$, let $b(k) = \lfloor r_k / 2^n \rfloor \in \{0, 1\}$ be the n -th bit and $s_k = r_k \bmod 2^n$ the low part, so $r_k = b(k)2^n + s_k$. Let

$$\gamma(k) = \lfloor (s_{k-1} + s_{k-2}) / 2^n \rfloor \in \{0, 1\}$$

be the carry into position n when $F(k-1)$ and $F(k-2)$ are added, and put

$$A(x) = \sum_{k=0}^{L-1} b(k)x^k, \quad \widehat{C}(x) = \sum_{k=0}^{L/2-1} \gamma(k)x^k.$$

A is the polynomial attached to $a(n)$.

3 The identity

Lemma 1. *For $m \geq 1$ the sequence $(F(k) \bmod 2^m)_{k \geq 0}$ is purely periodic with least period $3 \cdot 2^{m-1}$.*

Proof. Pure periodicity holds because $F(k-2) = F(k) - F(k-1)$ makes the recurrence invertible modulo any m ; the value $\pi(2^m) = 3 \cdot 2^{m-1}$ of the Pisano period is classical, see Wall [1]. $\square$

Thus $(r_k)_k$ has period L , so b, s, γ are functions on $\mathbb{Z}/L\mathbb{Z}$ and $r_k \equiv r_{k-1} + r_{k-2} \pmod{2^{n+1}}$ holds for every $k \in \mathbb{Z}/L\mathbb{Z}$.

Lemma 2. *In $\text{GF}(2)[x]$ there is an M with*

$$q(x)A(x) = \widehat{C}(x)(x^3 + 1)^{2^{n-1}} + M(x)(x^3 + 1)^{2^n}. \quad (2)$$

Proof. Splitting each summand into high bit and low part,

$$r_{k-1} + r_{k-2} = (b(k-1) + b(k-2) + \gamma(k))2^n + ((s_{k-1} + s_{k-2}) \bmod 2^n),$$

which is legitimate since $s_{k-1} + s_{k-2} < 2^{n+1}$. Reducing modulo 2^{n+1} and comparing with $r_k = b(k)2^n + s_k$, where $s_k < 2^n$, gives

$$b(k) \equiv b(k-1) + b(k-2) + \gamma(k) \pmod{2} \quad (k \in \mathbb{Z}/L\mathbb{Z}).$$

Multiplying by x^k and summing over $\mathbb{Z}/L\mathbb{Z}$, where the shift $k \mapsto k-1$ becomes multiplication by x modulo $x^L + 1$, yields $q(x)A(x) \equiv \Gamma(x) \pmod{x^L + 1}$ with $\Gamma(x) = \sum_{k=0}^{L-1} \gamma(k)x^k$. Now $\gamma(k)$ is a function of s_{k-1}, s_{k-2} alone, and $s_j = F(j) \bmod 2^n$ has period $3 \cdot 2^{n-1} = L/2$ by Lemma 1 applied with $m = n$. So the coefficient list of Γ is two copies of that of $\widehat{C}$:

$$\Gamma(x) = \widehat{C}(x)(1 + x^{L/2}) = \widehat{C}(x)(x^3 + 1)^{2^{n-1}}$$

by (1). Since $x^L + 1 = (x^3 + 1)^{2^n}$, also by (1), this is (2). $\square$

4 The theorem

Theorem 3. *For every $n \geq 1$, $v_{x+1}(A) \geq 2^{n-1}$. Consequently $(x^3 + 1)^{2^{n-1}-1}(x + 1)$ divides A .*

Proof. Both terms on the right of (2) are divisible by $(x^3 + 1)^{2^{n-1}}$ and $v_{x+1}((x^3 + 1)^m) = m$, so $v_{x+1}(qA) \geq 2^{n-1}$. Since $q(1) = 1$ we have $v_{x+1}(q) = 0$, hence $v_{x+1}(A) = v_{x+1}(qA) \geq 2^{n-1}$.

The same equation read with v_q gives $v_q(qA) \geq 2^{n-1}$, and $v_q(q) = 1$, so $v_q(A) \geq 2^{n-1} - 1$ (this is Conjecture 1). Writing $(x^3 + 1)^{2^{n-1}-1}(x + 1) = (x + 1)^{2^{n-1}} q^{2^{n-1}-1}$ and comparing exponents, $v_{x+1}(A) \geq 2^{n-1}$ and $v_q(A) \geq 2^{n-1} - 1$ are exactly what is needed. $\square$

Corollary 4. *For every $n \geq 1$, $a(n)$ is divisible in $\text{GF}(2)[X]$ by $(x^3 + 1)^{2^{n-1}-1}(x + 1)$. This is Karttunen's Conjecture 2.*

Remark 5 (honest accounting). Once the identity of Lemma 2 is in hand, Conjecture 1 and Conjecture 2 are the two valuations one can read off it, and Conjecture 2 costs one extra line: it is what the $(x + 1)$ -side gives when no power is lost to the cancellation. The entry states them as two separate conjectures, and what is settled here is the second.

Remark 6 (sharpness). Equation (2) gives the exact value $v_{x+1}(A) = 2^{n-1} + v_{x+1}(\widehat{C})$ whenever $v_{x+1}(\widehat{C}) < 2^{n-1}$. Computation gives $v_{x+1}(\widehat{C}) = 0$ for $n = 1$ and for $4 \leq n \leq 12$, so the bound of Theorem 3 is attained there; at $n = 2$ and $n = 3$ one has $v_{x+1}(\widehat{C}) = 1$ and 3 , so $v_{x+1}(A) = 3$ and 7 rather than 2 and 4 . Whether $v_{x+1}(\widehat{C}) = 0$ for all $n \geq 4$ — equivalently whether the number of carries in one period of length $L/2$ is odd — is not proved here.

5 Verification

A verification script, written separately from this note, checks every assertion and prints every number quoted. It (i) recomputes $a(0), \dots, a(3)$ and matches the published 6, 24, 1440, 5728448; (ii) checks the carry identity of Lemma 2 at all L residues for $n = 0, \dots, 16$, that is up to $L = 196608$, with no failure; (iii) checks the half-period property of γ over the same range, with no failure; (iv) verifies (2) by explicit polynomial division for $n = 1, \dots, 12$; (v) computes $v_{x+1}(A)$, obtaining 2^{n-1} for $n = 1$ and $4 \leq n \leq 12$ and the exceptional 3 and 7 at $n = 2, 3$, together with $v_{x+1}(\widehat{C}) = 0, 1, 3, 0, \dots, 0$; and (vi) divides $a(n)$ by $(x^3 + 1)^{2^{n-1}-1}(x + 1)$ directly for $n = 1, \dots, 12$, obtaining remainder zero in every case.

References

- [1] D. D. Wall, *Fibonacci series modulo m* , Amer. Math. Monthly 67 (1960), 525–532.
- [2] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.
- [3] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A036284>, entry A036284, consulted 23 August 2026.